

SHASHWAT SAXENA

DATA ENGINEER

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PROFESSIONAL SUMMARY

Results-driven **Data Engineer with 4+ years of experience** architecting scalable ETL data pipelines, modernizing data architectures, and deploying cloud-native big data solutions. Proficient in Python, SQL, Apache Spark, and cloud ecosystems (AWS & Azure). Proven track record of improving system uptime, reducing operational costs, and transforming raw data into actionable business intelligence.

Core Expertise: Specialized in orchestrating automated big data processing pipelines that drastically enhance data pipeline accuracy, system reliability, and overall operational efficiency.

SKILLS & EXPERTISE

Big Data & Processing

- Apache PySpark
- Apache Kafka
- Databricks
- Apache Iceberg
- dbt (Data Build Tool)
- Hadoop / HDFS
- MapReduce
- Hive

Cloud Technologies

- AWS (S3, EMR, Redshift)
- AWS Glue & Athena
- AWS Lambda
- Azure Data Factory
- Azure Synapse
- Azure Databricks

Databases & Warehousing

- Snowflake
- SQL & NoSQL
- Oracle
- MongoDB
- Microsoft SQL Server
- MySQL

Programming & Foundations

- Python
- Data Structures
- Algorithms
- System Design
- Data Modeling

GenAI & Applied AI

- Generative AI
- RAG Architecture
- Vector Databases
- Prompt Engineering
- LLMOps

Monitoring & Orchestration

- Apache Airflow
- AWS CloudWatch
- Grafana
- Prometheus

DevOps & CI/CD

- Terraform (IaC)
- Docker
- Kubernetes
- Jenkins
- Git (GitHub)
- Bitbucket
- Octopus
- Bamboo
- Linux / Unix

Data Science Libraries

- Pandas
- NumPy
- Scikit-Learn
- Boto3 (AWS SDK)

Concepts & Methodologies

- ETL / ELT Pipelines
- Data Warehousing (EDW)
- Agile & Scrum
- Event-Driven Architecture

Data Governance & Security

- Data Cataloging
- Data Lineage
- IAM & RBAC
- Data Quality

Professional Skills

- Problem Solving
- Technical Communication
- Cross-functional Teamwork
- Critical Thinking

PROFESSIONAL EXPERIENCE



Data Engineer

IBM (Client: Barclays)

07/2024 - Present

- Optimized ETL Data Pipelines:** Engineered and streamlined data processing workflows using **AWS Glue**, significantly improving data quality and achieving a **25% reduction** in operational costs.
- Performance Optimization:** Refactored and optimized **PySpark** scripts, accelerating data processing speed by **25%** and saving over 200 hours of compute time annually.
- Real-time Monitoring & Reliability:** Architected robust streaming and monitoring solutions utilizing **Apache Kafka** and PySpark, boosting overall pipeline reliability by **40%** and maintaining a **99.9%** system uptime.
- Cloud Infrastructure Enhancements:** Upgraded cloud architecture using native AWS services, successfully reducing system downtime by **30%** and cutting infrastructure costs by 15%.
- Generative AI & LLMs:** Designed and deployed a scalable **RAG (Retrieval-Augmented Generation)** pipeline using LangChain, Vector DBs, and OpenAI API to automate complex financial document summarization, reducing manual review time by 40%.



Software Engineer

Pine Labs Private Limited

02/2021 - 07/2024

- Cloud Data Lake Migration:** Spearheaded the strategic migration from traditional RDBMS to a modern Data Lake architecture on **Azure Data Lake / HDFS**, unlocking advanced scalable analytics.
- Big Data Processing:** Led the end-to-end migration of legacy data workloads to **Azure Databricks**. Optimized PySpark data pipelines, achieving a **10x reduction** in execution time while processing over 5 million daily records.
- Data Warehouse Implementation:** Designed and deployed a 100GB+ enterprise Data Warehouse for a leading airline client using **Azure Synapse, Data Factory, and SQL Server**, consolidating feeds from 10+ distinct vendors.
- Stream Processing Architecture:** Developed fault-tolerant, real-time data ingestion and transformation pipelines utilizing **Apache Spark Streaming** and **Apache Kafka** for high-velocity data environments.



Data Analyst Intern

Nuvoretail Enlytical Technologies Pvt Ltd

10/2020 - 01/2021

- Utilized distributed data analysis to optimize the online billing system, demonstrating a strong cost-benefit ratio that drove a **50% speed and security improvement**.
- Developed and deployed 8+ statistical models to forecast future sales trends, achieving **90% prediction accuracy** through robust distribution fitting techniques.

EDUCATION

Ph.D. Scholar - Trustworthy & Explainable AI (XAI)

Vellore Institute of Technology (VIT)

Pursuing

Expected 2029

Research Focus: Architecting cutting-edge methodologies to make Large Language Models (LLMs) transparent, bias-free, and ethically compliant for large-scale enterprise data systems.

Master of Computer Application (MCA)

Vellore Institute of Technology (VIT)

8.91 / 10

07/2019 - 07/2021

Bachelor of Computer Application (BCA)

Mahatma Jyotiba Phule Rohilkhand University

65%

08/2015 - 07/2018

CERTIFICATIONS

- AWS Certified Data Engineer - Associate**
- PCAP™ – Certified Associate Python Programmer**